Year: Fourth Year **Semester – VIII**

Course: Advanced Databases **Course Code: 17YCSE11**

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
3	0	0	3	10	20	10	10	--	50	--	100
Max. Time, End Semester Exam (Theory) -3Hrs.									**End Semester Exam (Lab) – 00 Hrs.**		

Prerequisites:

Database management Systems

Objectives:

1	To learn the basics of DBMS-files, commands, storage and structure
2	To understand modern database management system
3	To understand and apply advance query optimization
4	To learn and understand information retrieval and databases
5	To understand and apply the security and privacy

Unit No	Details	Hours
1	**Module 1:** Database Concepts (File System and DBMS), Database Storage Structures (Tablespace, Control files, Data files)	5
	Module 2: Structured and Unstructured data, SQL Commands (DDL, DML & DCL), Dataware Housing concept and tools (ETL tools),	4
2	**Module 1:** Introduction to Modern databases,NoSQL, NewSQL, NoSQLVs RDBMS databases, Advantages &Tradeoffs, Working with MongoDb,No,SQL, Data Models , XML, working with MongoDB),	5
	Module 2: Tools , OLTP and OLAP, data preparation and cleaning techniques	4
3	**Module 1:** Advanced query optimization: Volcano/Cascades framework for query optimization; multi-query optimization, materialized views and view maintenance	5
	Module 2: index and view selection, database tuning. Adaptive query processing and optimization	4
4	**Module 1:** Query processing on RDF data, Transaction and query processing on main-memory and columnar databases.	5
	Module 2: Data streams and stream management systems. Information retrieval and databases	4
5	**Module 1:** Handling uncertain and precise data. Security and privacy.	5
	Module 2: Crowd-sourced databases, applications of declarative querying outside of database applications	4

Outcomes:

On completion of the course, learners will be able to–

1	Explain DBMS-files, commands, Database storage and structure
2	Discuss modern database management system
3	Explain and apply advance query optimization

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4	Discuss information retrieval and databases
5	Explain and apply the security and privacy

Text Books

1. Abraham Silberschatz, Henry F. Korth and S. Sudarshan, Database System Concepts 6th Ed, McGraw Hill, 2010.
2. Transaction Processing, Concepts and Techniques, J. Gray and A. Reuter, Morgan Kauffman, 1994

Reference Books

1. MongoDB in Action by Kyle Banker
2. The Definitive Guide –MongoDB by Kristina Chodorow
3. MongoDB Aggregation Framework Principles and Examples by John Lynn
4. Getting Started with NoSQL by GauravVaish
5. Database System Concept by Henry Korth, S.Sudarshan& Abraham Silberschatz